

Atomic structure

Heisenberg's uncertainty principle

$$\Delta x \Delta p \geq \frac{h}{4\pi}$$

Lyman- $n=1 \rightarrow UV$
 Balmer- $n=2 \rightarrow \text{visible}$
 Paschen- $n=3$
 Brackett- $n=4$ } infra Red
 Pfund- $n=5$

Hydrogen spectrum

Rydberg's Constant

$$\frac{1}{\lambda} = \bar{\nu} = R_H \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right] Z^2$$

first line = $n+1$

last line $\rightarrow n_2 = \infty$

higher stage $\rightarrow$ ground state

$\rightarrow$ from $n_2 \rightarrow n_1$

$$\frac{(n_2 - n_1)(n_2 - n_1 + 1)}{2}$$

spectral lines
 $\rightarrow n_2 - n_1$

DeBroglie equation

$$\lambda = \frac{h}{mv} \quad \lambda = \sqrt{\frac{150}{v}} \text{ \AA}$$

$$\lambda = \frac{h}{\sqrt{2mK_E}}$$

Bohr's atomic model

$$\rightarrow F_c = \frac{mv^2}{r} = \frac{KZe^2}{r^2}$$

$$r_1 = 0.529 \times \frac{n^2}{Z} \text{ \AA}$$

$$v = 2.188 \times 10^6 \frac{Z}{n}$$

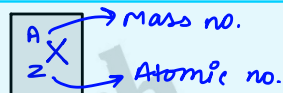
$$E = -13.6 \frac{Z^2}{n^2}$$

$$T.E. = -K.E$$

$$P.E. = -2K.E$$

$$T.E. = \frac{P.E}{2}$$

$$\Delta E_n = 13.6 \times Z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$



Isotopes: - same atomic no.

eg - ${}^6C^{12}$, ${}^6C^{13}$, ${}^6C^{14}$

Isodiaphers: - same difference in no. of protons & neutrons

eg - ${}^5B^{11}$, ${}^6C^{13}$

Isosters: - same no. of atoms & e^- .

eg - CO_2 , N_2O

Isotones: - same no. of neutrons.

eg - ${}^1H^3$, ${}^2He^4$

Isoelectronic: - same no. of e^-

eg - Ca^{2+} , Ar , S^{2-}

Aufbau principle

electrons are filled in increasing order of energy of subshells

$n + l$ rule

smaller value of n ;
smaller energy level

Pauli's exclusion principle

Two e^- can't have all same quantum no.

Hund's rule

e^- are distributed to have max. no. of unpaired electrons.

Principal quantum no. (by bohr)

$$n = 1, 2, 3, 4, \dots \quad \text{Total no. of } e^- \text{ subshell} = 2n^2$$
$$L = \frac{nh}{2\pi} \quad \text{Total no. of orbitals in a shell} = n^2$$
$$\text{Total no. of subshell} = n$$

Quantum no.

Spin quantum number

→ by Goudsmith

$$\rightarrow S = \pm \frac{1}{2}$$

spin momentum

$$\sqrt{S(S+1)} \frac{h}{2\pi}$$

Magnetic quantum no.

→ by Linde

$$\Rightarrow m \Rightarrow -l \text{ to } +l$$

for $p \rightarrow m = 3$

$d \rightarrow m = 5$

Azimuthal/angular quantum no. (l)

→ also called secondary/
subsidiary no.

→ By Sommerfeld

$$s \rightarrow l = 0$$

$$p \rightarrow l = 1$$

$$d \rightarrow l = 2$$

$$f \rightarrow l = 3$$

$$\text{no. of } e^- \text{ in subshell} = 2(2l+1)$$

Orbital angular momentum

$$\sqrt{l(l+1)} \frac{h}{2\pi}$$